# Nazar Blanco Kataran

Junior Software Developer · Quantum Computing

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## PROFESSIONAL SUMMARY

Multimedia engineer with a master's degree in Quantum Computing and hands-on experience in backend development with Go, web applications with React, and quantum algorithms with Qiskit. For my master's thesis, I designed a 10-qubit circuit that generates procedural maps from quantum states and renders them in a 3D environment in Unity. I have worked in English-speaking environments in Germany and Ireland, and I bring a solid foundation in visual design. I am seeking a junior position in software development, front-end development, or quantum engineering.

## PROFESSIONAL EXPERIENCE

**Research Intern — AIR Institute** 06/2025 – 10/2025

*Castilla y León, Spain (Remote)*

- I researched the preparation of quantum states using the Grover-Rudolph and Möttönen algorithms and their application to procedural graphics generation—a line of research that became the focus of my master's thesis.
- I implemented and validated circuits of up to 10 qubits on simulators (Google Colab and IBM Quantum Composer) using Qiskit, and analyzed how simulation costs increase as qubits are added.
- Together with my advisor, I wrote a draft scientific article in LaTeX (Overleaf) based on the results of my master's thesis.
- I submitted the documented code and a technical report with the results of each method.

**Digital Marketing Assistant — La Jijonenca** 11/2023 – 10/2024

*Jijona, Spain*

- I managed the brand's Instagram account: planning and creating weekly posts (photography, captions, and hashtags).
- I designed the signage and graphic elements requested by the ice cream shops in the network, maintaining the brand's visual identity.
- I updated the images and banners on the corporate website.
- Prepared and retouched the images and banners for the Christmas campaign on the turrón page using Photoshop.

**Backend Developer Intern — Geoffrey Emmans** 07/2023 – 09/2023

*Cologne, Germany*

- I designed and developed an API in Go to manage clients in sorted lists with different roles, based on a relational database.
- I implemented role-based access control: superadministrators could create, edit, and delete clients, while others could only view them.
- I validated the logic, API calls, and data model with small-scale tests as a preliminary step before integrating it into the company's backend.
- I developed the project independently, under the supervision of the technical lead, in English, and using version control on GitHub.

## PROJECTS

**q-state-prep — efficient preparation of quantum states** 11/2026

*github.com/nazarbk/q-state-prep*

- A Python library that prepares quantum states to generate procedural noise maps and compares the exact preparation with variational circuits designed for current hardware (NISQ).
- Using a variational circuit trained with COBYLA, I reduced the number of CNOT gates from about 578 to 20, while maintaining a fidelity of over 95%; branch pruning reduces them by 20% with a fidelity of over 99%.

- It includes an interactive explorer in Streamlit with plots showing accuracy, cost versus accuracy, and convergence, as well as tests using Pytest.
  *Tools: Python, Qiskit, SciPy, NumPy, Streamlit, Pytest*

**Procedural Map Generation Using Quantum Algorithms (Master's Thesis)** 10/2025

*https://colab.research.google.com/drive/1KA3h1YA677x-WFca_UtbINorLCKW7Cdd*

- I designed a 10-qubit circuit (two subcircuits of 3 data qubits and 2 ancilla qubits, one per axis) that combines Grover-Rudolph state preparation with QFT and IQFT to smooth the distribution.
- I converted the 1,024-value histogram into a 32×32 heightmap with smooth transitions and imported it into Unity using a C# script that generates the 3D terrain.
  *Tools: Qiskit, Python, NumPy, Unity, C#*

**BB84 Protocol Simulator** 06/2025

*bb84-protocol.streamlit.app*

- It simulates quantum key distribution, detects a spy's interference, and displays each step on the Bloch sphere.
  *Tools: Python, Qiskit, Streamlit, Matplotlib*

**Qonverse (hackathon)** 05/2025

*qonverse.vercel.app*

- AI conversation simulator for practicing communication in different roles, featuring authentication, a history log, and a free plan with usage limits.
  *Tools: React, TypeScript, Firebase Firestore, Gemini API, Clerk*

**Quantum Computing Interactive Explorer** 03/2025

*bit-vs-qubit.vercel.app*

- Bilingual educational app with six interactive modules—from qubits to entanglement—and a BB84 cryptography minigame.
  *Tools: React 18, Vite, Framer Motion, KaTeX*

## EDUCATION

**Master's Degree in Quantum Computing** 2024 – 2025

*UNIR*

**Bachelor's Degree in Multimedia Engineering** 2018 – 2024

*Universidad de Alicante*

## SKILLS

Python · Go · JavaScript · TypeScript · C++ · C# · Java · React · Angular · Ionic · HTML · CSS · Tailwind · Framer Motion · Vite · Node.js · APIs REST · MongoDB · Firebase Firestore · Qiskit · IBM Quantum · PennyLane · QFT · Git · GitHub · Google Colab · Streamlit · Unity · LaTeX · Vercel · Photoshop · Illustrator · Canva · API REST · NumPy

## LANGUAGES

Spanish: Native (C2) · English: Advanced (B2)

## CERTIFICATIONS

- AI-assisted Software Engineering for Quantum Computing and PQC — QuLearnLabs, 2025 (EQF 4–6)
- Quantum Computing in Practice — IBM, 2026
- Using a Quantum Computer — IBM, 2026
- Basics of Quantum Information (badge) — IBM, 2026
- Quantum Business Foundations (badge) — IBM, 2026